

NAME: _____

PERIOD: _____

Two Displays, One Data Set

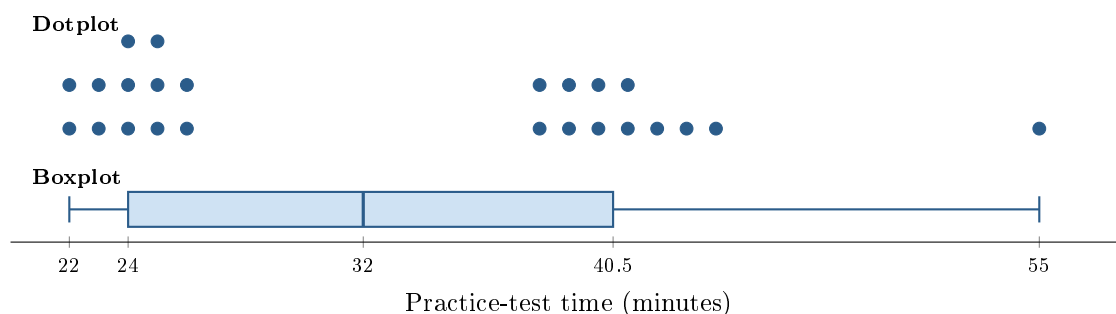
AP Statistics · Unit 1 · Topic 1.8

[STAR] THE PROBLEM

The displays show practice-test times, in minutes, for 24 students. Two students interpret the boxplot.

Omar: “Most students took about 32 minutes—the middle of the box.”

Nia: “A boxplot cannot tell us whether most students took about 32 minutes.”



FIRST TAKE — one sentence before you work the parts

Who do you trust more, Omar or Nia? Point to one feature in either display.

[BOOK] DOTPLOTS AND BOXPLOTS (keep on the page)

A **dotplot** keeps every observation, so it can show clusters, gaps, peaks, and isolated values. A **boxplot** compresses the data to the five-number summary: minimum, Q1, median, Q3, maximum (with separately marked outliers). The box contains the middle 50% of observations. A boxplot shows center and spread efficiently, but it may hide modality, clusters, and gaps.

Work (a)–(c).

DOK 1 (a) Read the median and the two quartiles from the boxplot.

DOK 2 (b) In two sentences, describe the dotplot's groups and empty interval. Count how many students took more than 30 minutes.

DOK 3 (c) Whose interpretation is better supported? Use a dotplot feature to explain what the boxplot hides. Choose a display for deciding whether more than half the students need over 30 minutes, and defend your choice.

Frame: _____ is better supported because the dotplot shows _____ around the median of 32 minutes.

Frame: The boxplot shows _____, but it hides _____, so the coach should use the _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.